

Basic Information

Experiment No.: 06

Date: 24-09-2026

Title: Java ListIterator Interface using ArrayList

Objective / Aim

To implement the ListIterator interface to traverse an "ArrayList" in both forward and backward directions.

System Requirements

Hardware

- Computer/Laptop
- Minimum 4 GB RAM

Software

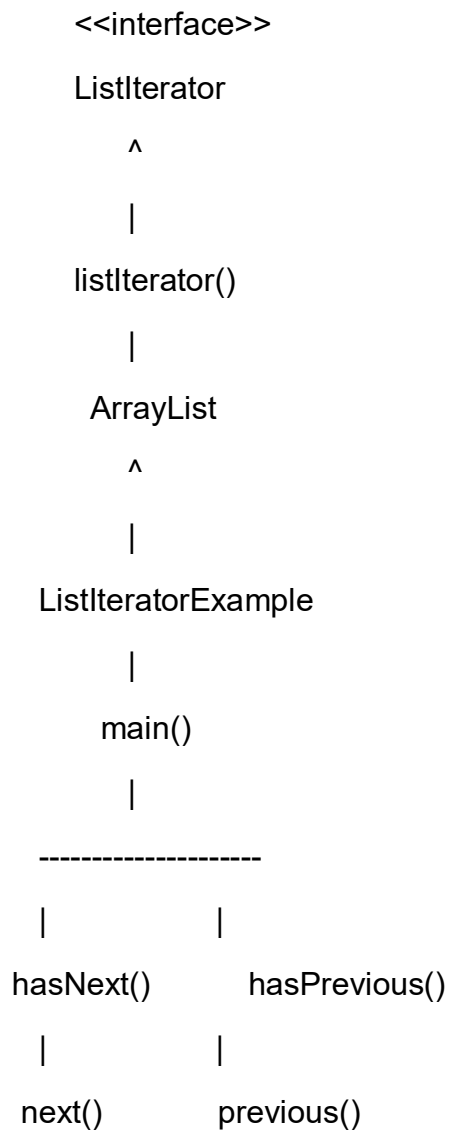
- Java JDK
 - Any Java IDE or Command Prompt
 - Operating System: Windows/Linux/macOS
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Algorithm / Logic Description

1. Create an "ArrayList" and add elements.
2. Create a "ListIterator" using "list.listIterator()".
3. Traverse the list in the forward direction using "hasNext()" and "next()".
4. Display each element.
5. Traverse the list backward using "hasPrevious()" and "previous()".
6. Display each element.

7. End the program.

UML Class Diagram



Source Code

```
package collectionframework;
```

```
import java.util.List;
import java.util.ArrayList;
import java.util.ListIterator;

public class ListIteratorExample {
    public static void main(String[] args) {

        List<String> list = new ArrayList<>();

        list.add("Apple");
        list.add("Banana");
        list.add("Mango");

        ListIterator<String> it = list.listIterator();

        // Forward direction
        System.out.println("Forward:");

        while (it.hasNext()) {

            System.out.println(it.next());
        }

        // Backward direction
        System.out.println("\nBackward:");

        while (it.hasPrevious()) {
```

```
        System.out.println(it.previous());
    }
}
}
```

Compilation & Execution Commands

Save the file as:

ListIteratorExample.java

Since the program contains:

package collectionframework;

the file should be inside the "collectionframework" folder.

Compile:

javac ListIteratorExample.java

Run:

java collectionframework.ListIteratorExample

Output Screen

Forward:

Apple

Banana

Mango

Backward:

Mango

Banana

Apple

Result

The ListIterator interface was successfully implemented to traverse the "ArrayList" in both forward and backward directions using "hasNext()", "next()", "hasPrevious()", and "previous()".